

Overview: Today's Milestones

1. **System Performance:** Multi-threaded Asynchronous Engine.
2. **Signal Robustness:** Signal Quality Index (SQI) & Hybrid SpO2.
3. **Tracking Precision:** EMA-Smoothed Head Pose & 3D Visualization.
4. **Interactive Logic:** Hybrid Fatigue Decision Engine.
5. **UI/UX:** Professional side-panel Dashboard.
6. **Validation:** Automated Benchmarking (UBFC-rPPG).

1. System Performance

Asynchronous Inference Engine

- **The Problem:** UI "stuttering" during heavy AI model calculations.
- **The Solution:** Decoupled AI models into a background `worker` thread.
- **Result:**
 - Smooth **30 FPS** camera feed.
 - Zero UI lag during Heart Rate and Fatigue analysis.
 - Improved thread safety using `Queue` and `threading`.

2. Physiological Robustness

Filtering the Noise

- **Signal Quality Index (SQI):**
 - Real-time FFT analysis to calculate **SNR (Signal-to-Noise Ratio)**.
 - Automatically displays "**Searching...**" if lighting is poor.
- **Hybrid SpO2 Logic:**
 - Blends AI Deep Learning with traditional analytical formulas.
 - **Outcome:** Vitals stabilized from erratic 80% to a healthy **95-98% range**.

3. Precise Head Pose Tracking

3D Directional Awareness

- **Geometric Correction:** Fixed 3D mapping to resolve axis flipping.
- **EMA Smoothing:** Integrated Exponential Moving Average filters to eliminate jitter.
- **3D Visualization:**
 - Real-time **RGB 3D Coordinate Axis** on the nose.
 - Instant visual feedback for Pitch (Nod), Yaw (Turn), and Roll.

4. Fatigue Decision Engine

Interactive Intelligence

- **Moving beyond Placeholders:** Combined AI predictions with physical heuristics.
- **Real-Time Responsiveness:**
 - **PERCLOS:** Monitors percentage of eye closure.
 - **Yawn Detection:** Instant trigger for "Critical" state alerts.
- **Hybrid Blend:** Blends BiLSTM temporal trends with rule-based safety logic (80/20 split).

5. Professional Dashboard

UI/UX Transformation

- **Dashboard Layout:**
 - Dark-themed side panel for high visibility.
 - **Real-Time Pulse Wave:** ECG-style scrolling graph.
 - **Fatigue Meter:** Color-coded (Green -> Yellow -> Red).
- **Vitals Cards:** Professional data readouts for HR and SpO2.
- **Layout Logic:** Compact design compatible with standard 480p/720p webcams.

6. Benchmarking & Validation

Data-Driven Results

- **Dataset:** Validated against **UBFC-rPPG** (8 subjects).
- **Findings:**
 - **HR MAE: 6.03 BPM** (Excellent performance for a standard webcam).
 - **SpO2 MAE: 17.4%** (Area identified for future calibration).
- **Automation:** Created `benchmark_rppg.py` for continuous accuracy testing.

Conclusion & Next Steps

- [] **Blink Detection:** Monitor frequency and duration.
- [] **Edge Optimization:** Convert `.h5` models to **TensorFlow Lite**.
- [] **Audio Alerts:** Implement voice/beep warnings for Critical Fatigue.

Questions?

AI Driving Monitoring System

Technically Robust. Visually Professional.